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WT : EXPERIMENT 5

Aim : Simulation of an Automatic Water Dispensing System using Tinkercad

Theory :

The concept of an Automatic Water Dispensing System involves the use of embedded systems and sensors to detect a glass and dispense water without manual intervention. In this project, we will use an Arduino Uno, an ultrasonic sensor (HC-SR04) to detect the presence of a glass, and a water pump (simulated with an LED or servo motor in Tinkercad) to control water flow. The system will be simulated using Tinkercad, where we will model the components, make connections, and test the functionality.

Working Principle:

The system operates as follows:

1. **Ultrasonic Sensor Detection:**
    - The ultrasonic sensor continuously monitors for the presence of a glass.
    - When a glass is placed within the defined range (e.g., **≤ 10 cm**), the system detects it.

2. **Activating the Water Pump:**
    - If the sensor detects a glass, the Arduino **activates the water pump** (or an LED in the simulation).
    - The pump stays on for a fixed duration (e.g., **3 seconds**) to dispense water.

3. **Stopping the Water Flow:**
    - If no glass is detected, the pump remains off.
    - The system continuously checks for the glass presence to automate the process.

Steps for Simulation: Automatic Water Dispenser using Tinkercad

1. **Build the Circuit**
    - Place the Components:
        - Place the Arduino Uno board in the Tinkercad workspace.
        - Drag the HC-SR04 ultrasonic sensor onto the workspace. This sensor will detect the presence of a glass under the water dispenser.
        - Drag an LED (to represent the pump in the simulation) onto the workspace.
2. **Make the Connections**
    - Connecting the Ultrasonic Sensor (HC-SR04) to Arduino:
        - Connect the VCC pin of the sensor to the 5V pin on the Arduino.
        - Connect the GND pin of the sensor to the GND pin on the Arduino.
        - Connect the Trig pin of the sensor to Pin 9 on the Arduino.
        - Connect the Echo pin of the sensor to Pin 10 on the Arduino.
    - Connecting the Water Pump (or LED) to Arduino:
        - Connect the IN pin of the water pump (or LED +) to Pin 7 on the Arduino.
        - Connect the GND pin of the water pump (or LED -) to the GND pin on the Arduino.
    - Jumper Wires:
        - Use jumper wires to connect components directly to the Arduino or through a breadboard.
3. **Test the System**
    - Once the circuit is set up and the code is uploaded to the Arduino, we can simulate the system in Tinkercad.
4. **Simulate the Circuit**
    - Click on "Start Simulation" in Tinkercad to begin the process.
    - The system will detect the presence of a glass using the ultrasonic sensor.
    - If a glass is detected, the water pump (represented by the LED) will turn on, simulating the dispensing of water.
5. **Adjust the Sensor Detection**
    - The ultrasonic sensor's detection range can be adjusted by modifying its trigger distance in the code.
    - Testing can be done by simulating different distances in Tinkercad to check when the pump turns on or off.

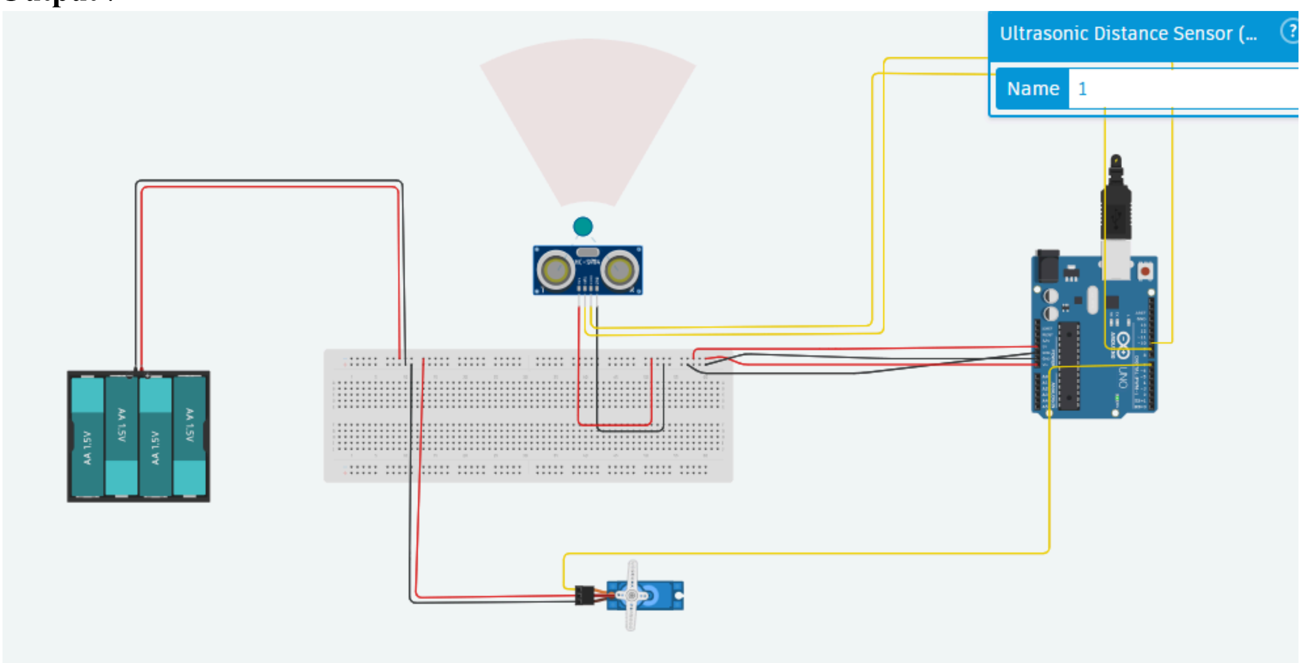
Code :

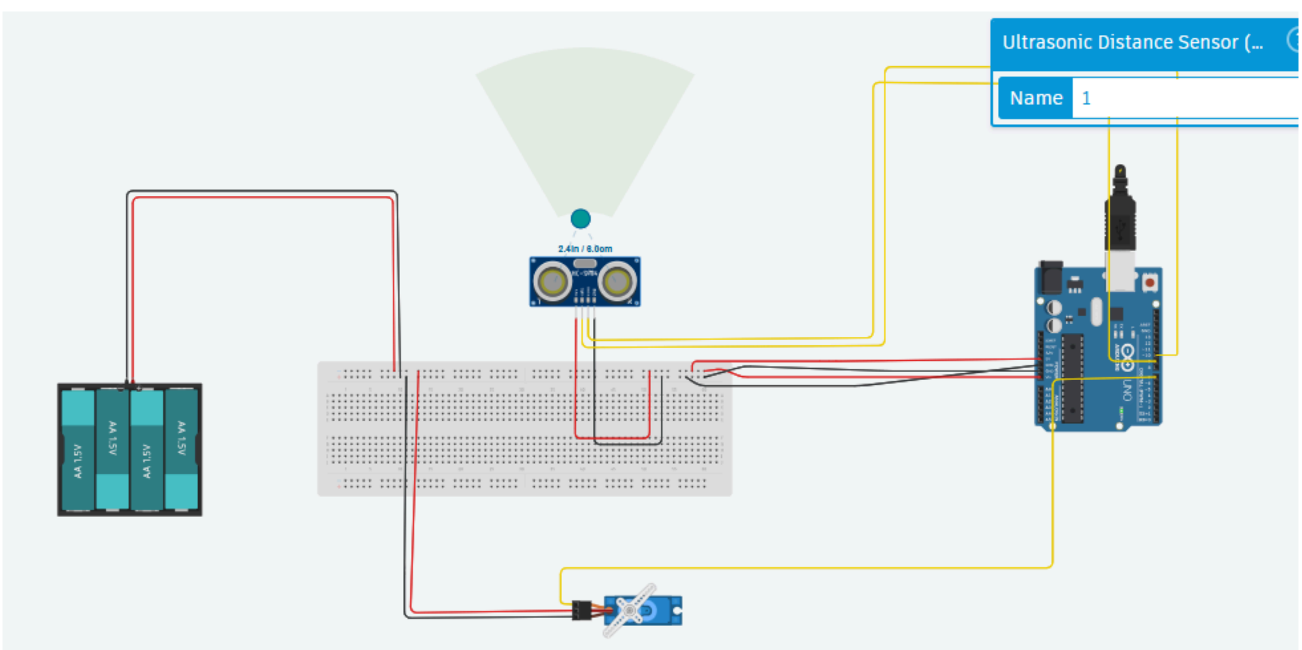
```
// Include necessary library
#define trigPin 10
#define echoPin 9
#define pumpPin 7 // Pin for the water pump (or LED in simulation)

void setup() {
```

```
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(pumpPin, OUTPUT);
  digitalWrite(pumpPin, LOW); // Ensure pump is off at start

  Serial.begin(9600);
}

void loop() {
  long duration;
  int distance;

  // Send pulse to trigger the ultrasonic sensor
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  // Read the echo pin and calculate distance
  duration = pulseIn(echoPin, HIGH);
  distance = duration * 0.034 / 2; // Convert time to distance (cm)

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  // If glass is detected within 10 cm, turn on the pump
  if (distance <= 10) {
    digitalWrite(pumpPin, HIGH); // Activate pump
    Serial.println("Glass detected. Pump ON.");
    delay(3000); // Dispense water for 3 seconds
    digitalWrite(pumpPin, LOW); // Stop pump
    Serial.println("Water dispensed. Pump OFF.");
  } else {
```

```
    digitalWrite(pumpPin, LOW); // Ensure pump stays off if no glass detected
    Serial.println("No glass detected. Pump OFF.");
  }

  delay(500); // Short delay to prevent excessive readings
}
```

Output :





Conclusion:

In this simulation, we created an Automatic Water Dispenser using Arduino and the HC-SR04 ultrasonic sensor. The system detects the presence of a glass under the dispenser using the ultrasonic sensor and activates the water pump (represented by an LED in the simulation) accordingly. The pump turns on when an object is detected within the specified range and turns off when the object is removed.

This simulation successfully demonstrates a touchless water dispensing system, which can be useful in places like public drinking stations, hospitals, and restaurants to maintain hygiene and reduce water wastage. The next step would be to integrate additional features such as a flow sensor for monitoring water usage or a relay module for controlling an actual water pump.